Loan Applicant Data - EDA Report

Overview

Overview | Alerts 10 | Reproduction

Alerts

`credit.policy` has constant value "1"	**Constant**
`fico` is highly overall correlated with `int.rate` and 1 other fields	High correlation
`int.rate` is highly overall correlated with `fico` and 1 other fields	High correlation
`revol.util` is highly overall correlated with `fico` and 1 other fields	High correlation
`delinq.2yrs` is highly imbalanced (63.7%)	Imbalance
`pub.rec` is highly imbalanced (67.3%)	Imbalance
`df_index` is uniformly distributed	Uniform
`df_index` has unique values	Unique
`dti` has 2 (2.0%) zeros	**Zeros**
`revol.util` has 3 (3.0%) zeros	**Zeros**

Variables

Select Columns

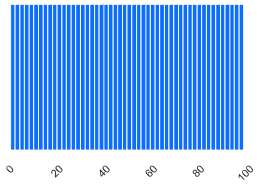
df_index

Real number (ℝ)

Uniform Unique

Distinct	100
Distinct (%)	100.0%
Missing	0
Missing (%)	0.0%
Infinite	0
Infinite (%)	0.0%
Mean	49.5

Minimum	0
Maximum	99
Zeros	1
Zeros (%)	1.0%
Negative	0
Negative (%)	0.0%
Memory size	932.0 I



More details

Statistics Histogram Common values

Extreme values

Quantile statistics

Minimum	0
5-th percentile	4.95
Q1	24.75
median	49.5
Q3	74.25
95-th percentile	94.05
Maximum	99
Range	99

Descriptive statistics

Standard deviation	29.011492
Coefficient of variation (CV)	0.58609075
Kurtosis	-1.2
Mean	49.5
Median Absolute Deviation (MAD)	25

Interquartile range (IQR)	49.5

Skewness	0
Sum	4950
Variance	841.66667
Monotonicity	Strictly increas

credit.policy

Categorical

Constant

Distinct	1
Distinct (%)	1.0%
Missing	0
Missing (%)	0.0%
Memory size	5.0 KiB

1 100

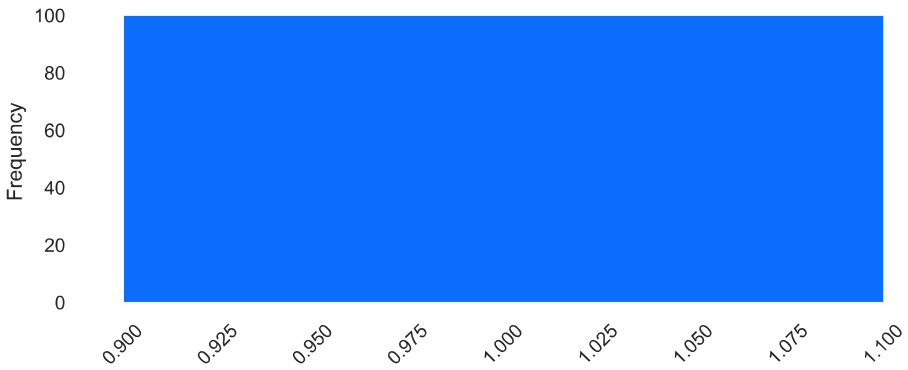
More details

Overview | Categories | Words | Characters

Common Values

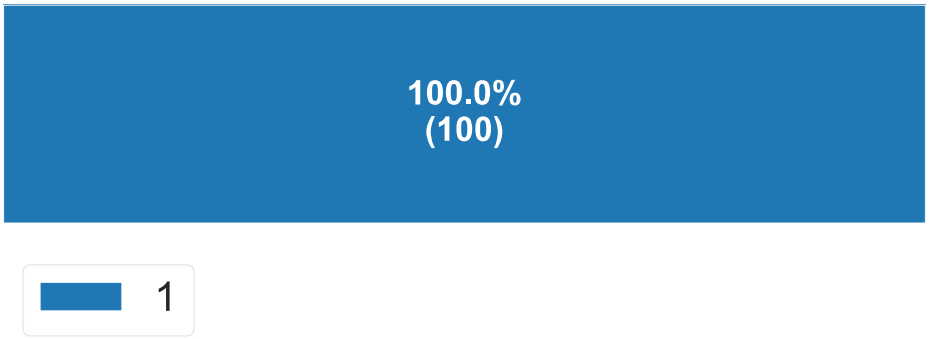
Value	Count	Frequency (%)
1	100	100.0%

Length



Histogram of lengths of the category

Common Values (Plot)



purpose

Categorical

Distinct	7
Distinct (%)	7.0%
Missing	0
Missing (%)	0.0%
Memory size	6.2 KiB

debt_co...	28
credit_ca...	23
all_other	23
home_i...	13
small_bu...	8
Other va...	5

More details

Overview | Categories | Words | Characters

Value	Count	Frequency (%)
debt_c…	28	28.0%
credit_c…	23	23.0%
all_other	23	23.0%
home_i…	13	13.0%
small_b…	8	8.0%
educati…	3	3.0%
major_…	2	2.0%

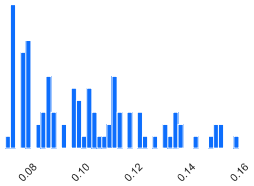
int.rate

Real number (ℝ)

High correlation

Distinct	46
Distinct (%)	46.0%
Missing	0
Missing (%)	0.0%
Infinite	0
Infinite (%)	0.0%
Mean	0.101028

Minimum	0.0714
Maximum	0.1596
Zeros	0
Zeros (%)	0.0%
Negative	0
Negative (%)	0.0%
Memory size	932.0 I



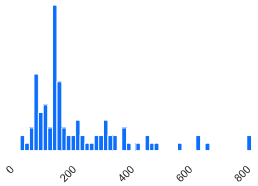
More details

installment

Real number (ℝ)

Distinct	83
Distinct (%)	83.0%
Missing	0
Missing (%)	0.0%
Infinite	0

Minimum	32.55
Maximum	829.1
Zeros	0
Zeros (%)	0.0%
Negative	0
Negative (%)	0.0%



Infinite (%)	0.0%
Mean	227.94

Memory size	932.0 I

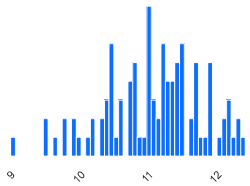
More details

log.annual.inc

Real number (ℝ)

Distinct	59
Distinct (%)	59.0%
Missing	0
Missing (%)	0.0%
Infinite	0
Infinite (%)	0.0%
Mean	11.079282

Minimum	8.9871
Maximum	12.429
Zeros	0
Zeros (%)	0.0%
Negative	0
Negative (%)	0.0%
Memory size	932.0 I



More details

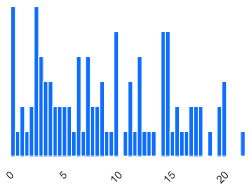
dti

Real number (ℝ)

Zeros

Distinct	91
Distinct (%)	91.0%
Missing	0
Missing (%)	0.0%
Infinite	0
Infinite (%)	0.0%
Mean	9.035

Minimum	0
Maximum	22.09
Zeros	2
Zeros (%)	2.0%
Negative	0
Negative (%)	0.0%
Memory size	932.0 I



More details

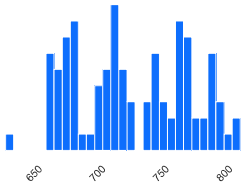
fico

Real number (ℝ)

High correlation

Distinct	29
Distinct (%)	29.0%
Missing	0
Missing (%)	0.0%
Infinite	0
Infinite (%)	0.0%
Mean	726.6

Minimum	627
Maximum	812
Zeros	0
Zeros (%)	0.0%
Negative	0
Negative (%)	0.0%
Memory size	932.0 I



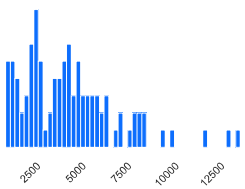
More details

days.with.cr.line

Real number (ℝ)

Distinct	91
Distinct (%)	91.0%
Missing	0
Missing (%)	0.0%
Infinite	0

Minimum	1110
Maximum	14008.
Zeros	0
Zeros (%)	0.0%
Negative	0
Negative (%)	0.0%



Infinite (%)	0.0%
Mean	4562.0937

Memory size	932.0

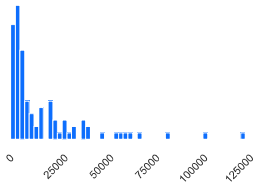
More details

revol.bal

Real number (ℝ)

Distinct	96
Distinct (%)	96.0%
Missing	0
Missing (%)	0.0%
Infinite	0
Infinite (%)	0.0%
Mean	17009.49

Minimum	0
Maximum	12800
Zeros	1
Zeros (%)	1.0%
Negative	0
Negative (%)	0.0%
Memory size	932.0



More details

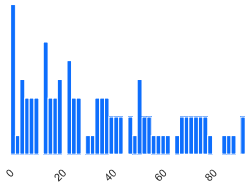
revol.util

Real number (ℝ)

High correlation **Zeros**

Distinct	93
Distinct (%)	93.0%
Missing	0
Missing (%)	0.0%
Infinite	0
Infinite (%)	0.0%
Mean	35.5

Minimum	0
Maximum	93.4
Zeros	3
Zeros (%)	3.0%
Negative	0
Negative (%)	0.0%
Memory size	932.0 I



More details

inq.last.6mths

Categorical

Distinct	5
Distinct (%)	5.0%
Missing	0
Missing (%)	0.0%
Memory size	5.0 KiB

0	55
1	24
2	11
3	9
4	1

More details

delinq.2yrs

Categorical

Imbalance

Distinct	3
Distinct (%)	3.0%
Missing	0
Missing (%)	0.0%
Memory size	5.0 KiB

0 89
1 9
2 2

More details

pub.rec

Categorical

Imbalance

Distinct	2
Distinct (%)	2.0%
Missing	0
Missing (%)	0.0%
Memory size	5.0 KiB

0 94
1 6

More details

not.fully.paid

Categorical

Distinct	2
Distinct (%)	2.0%
Missing	0
Missing (%)	0.0%

1 63
0 37

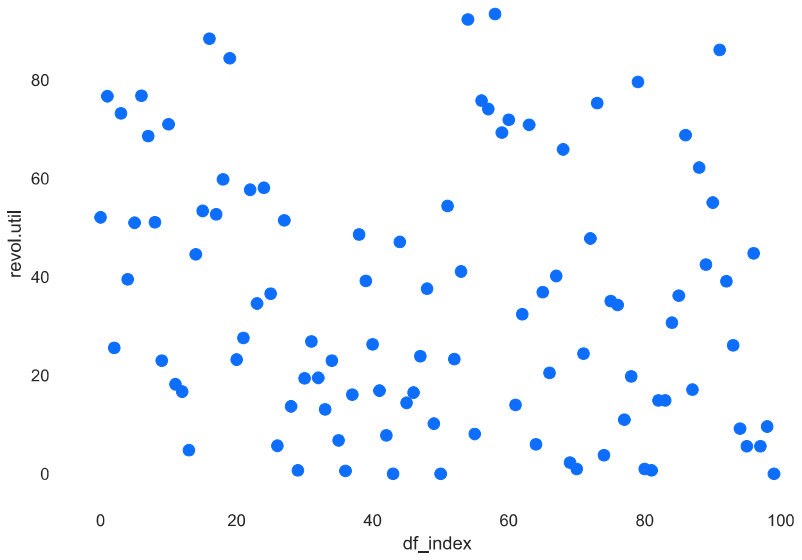
Memory size	5.0 KiB

More details

Interactions

df_index | int.rate | installment | log.annual.inc | dti | fico | days.with.cr.line | revol.bal | revol.util

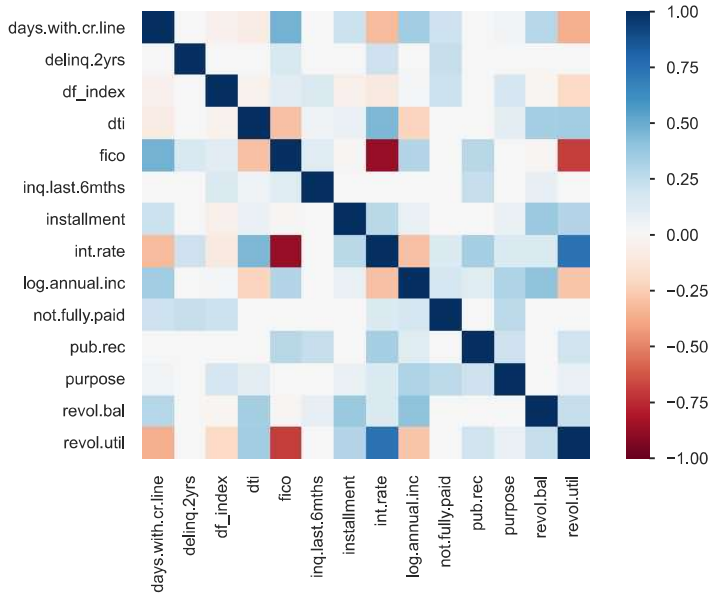
revol.util | df_index | int.rate | installment | log.annual.inc | dti | fico | days.with.cr.line | revol.bal



Correlations

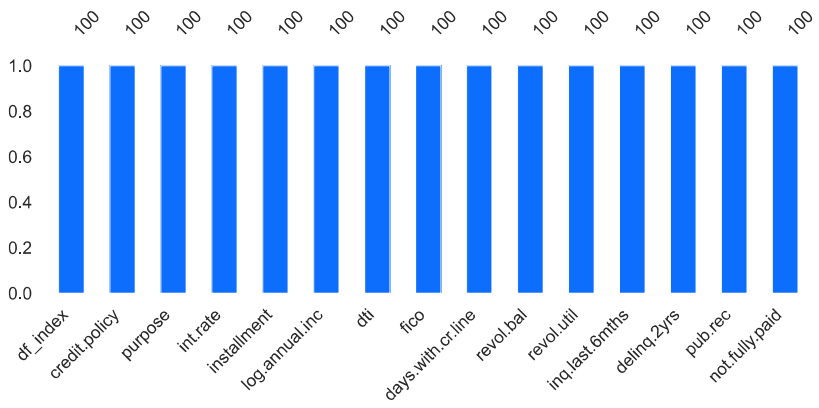
Auto

Heatmap Table



Missing values

Count Matrix



A simple visualization of nullity by column.

Sample

First rows | Last rows

	df_index	credit.policy	purpose	int.rate	insta
0	0	1	debt_consolidation	0.1189	829.
1	1	1	credit_card	0.1071	228.
2	2	1	debt_consolidation	0.1357	366.
3	3	1	debt_consolidation	0.1008	162.
4	4	1	credit_card	0.1426	102.
5	5	1	credit_card	0.0788	125.

	df_index	credit.policy	purpose	int.rate	insta
6	6	1	debt_consolidation	0.1496	194.(
7	7	1	all_other	0.1114	131.2
8	8	1	home_improvement	0.1134	87.19
9	9	1	debt_consolidation	0.1221	84.12

Report generated by YData.